M2S-03: Step 3 — Design: Create Target Architecture

> **Series:** M2S — Managed to SaaS Migration | **Notebook:** 3 of 9 | **Phase:** Plan | **Step:** Design | **Created:** March 2026 | **Last Updated:** 07/30/2026

With discovery and strategy complete, it's time to design the target architecture for your Dynatrace SaaS environment. This step produces the technical blueprints that guide every subsequent migration activity—network connectivity, ActiveGate topology, security controls, and high availability.

> **M2S Migration Journey — 3 Phases / 9 Steps**
>
> **Plan:** 1. Discover | 2. Strategize | **3. Design**
>
> **Upgrade:** 4. Prepare | 5. Execute | 6. Integrate
>
> **Run:** 7. Enable | 8. Expand | 9. Optimize

Table of Contents

Prerequisites

Before starting this notebook, you should have:

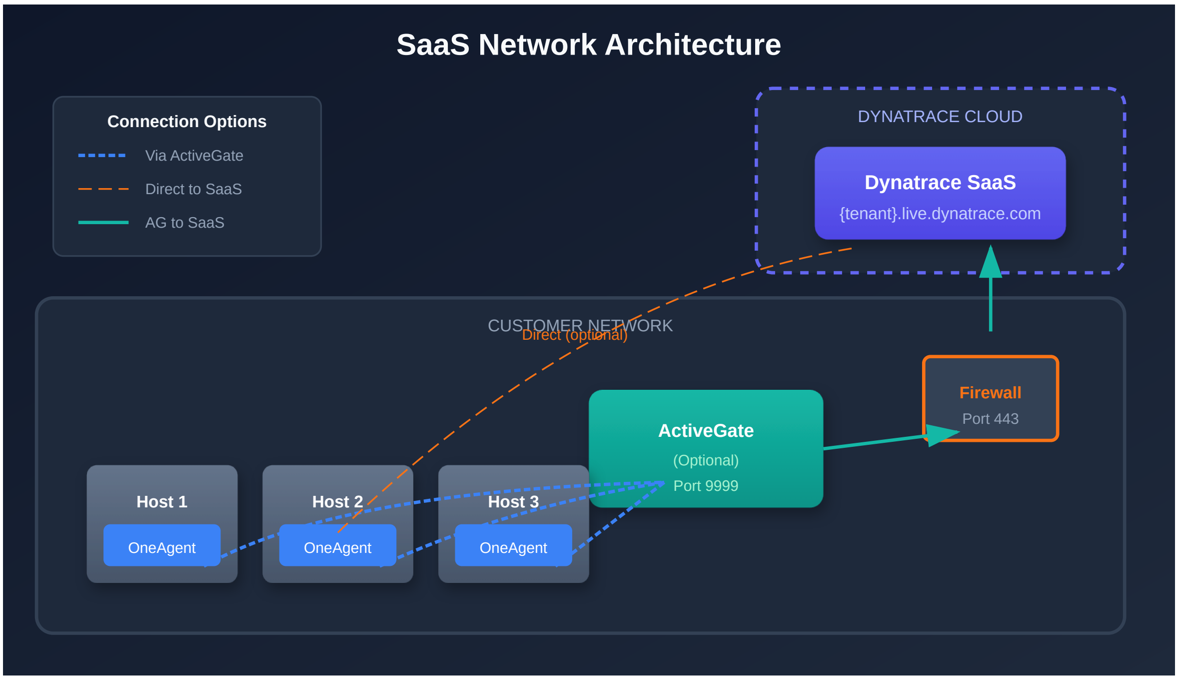
Requirement	Description
Steps 1–2 completed	Discovery inventory and migration strategy defined
SaaS tenant provisioned	Tenant ID and URL available
Network diagrams	Current Managed network topology documented
Firewall change process	Ability to request firewall rule changes
Security stakeholders engaged	Security and identity team involvement

> The Settings 2.0 schema for network zones (`builtin:networkzones.zones`) is catalogued alongside every other domain's schema in **AUTOM-02**'s consolidated Settings 2.0 Schema Catalog.

Learning Objectives

By the end of this notebook, you will:

- Design the network architecture for SaaS connectivity
- Plan network zone topology for optimized traffic routing
- Size and place ActiveGates for your environment
- Configure security controls including SSO and IAM
- Address high availability requirements
- Complete the architecture design checklist



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1. Network Architecture

The most significant architectural change when moving to SaaS is connectivity—your monitored infrastructure must reach Dynatrace SaaS endpoints over the internet.

1.1 SaaS Endpoints

All communication uses outbound HTTPS (port 443).

Endpoint Pattern	Purpose
`{tenant-id}.live.dynatrace.com`	Main SaaS cluster (agent communication, API)
`{tenant-id}.apps.dynatrace.com`	Apps and platform services

1.2 Connectivity Options

Option	OneAgent →	ActiveGate →	Best For
Direct	SaaS (443)	N/A	Open internet access
Via Proxy	Proxy → SaaS	N/A	Proxy-controlled environments

Option	OneAgent →	ActiveGate →	Best For
Via ActiveGate	ActiveGate (9999)	SaaS (443)	Restricted networks
Via AG + Proxy	ActiveGate (9999)	Proxy → SaaS (443)	Most restricted environments

> **Recommendation:** For environments where hosts cannot reach the internet directly, deploy ActiveGates as routing proxies. OneAgents connect to the ActiveGate on port 9999; the ActiveGate forwards traffic to SaaS on port 443.

1.3 Firewall Requirements

Minimum Requirements (Direct Connectivity):

Source	Destination	Port	Protocol
OneAgent hosts	`{tenant-id}.live.dynatrace.com`	443	HTTPS
OneAgent hosts	`{tenant-id}.apps.dynatrace.com`	443	HTTPS

ActiveGate Routing (Restricted Networks):

Source	Destination	Port	Protocol
OneAgent hosts	ActiveGate	9999	HTTPS
ActiveGate	`{tenant-id}.live.dynatrace.com`	443	HTTPS
ActiveGate	`{tenant-id}.apps.dynatrace.com`	443	HTTPS

1.4 DNS Resolution

Ensure DNS can resolve these domains from all hosts and ActiveGates:

- `{tenant-id}.live.dynatrace.com`
- `{tenant-id}.apps.dynatrace.com`
- `*.dynatrace.com` (for additional services and updates)

1.5 Connectivity Testing

Test connectivity from representative hosts before migration:

```
# Test HTTPS connectivity to SaaS
curl -v https://{tenant-id}.live.dynatrace.com/api/v1/time

# Test apps endpoint
curl -v https://{tenant-id}.apps.dynatrace.com

# Test from behind ActiveGate (verify AG is reachable)
curl -v https://{activegate-host}:9999
```

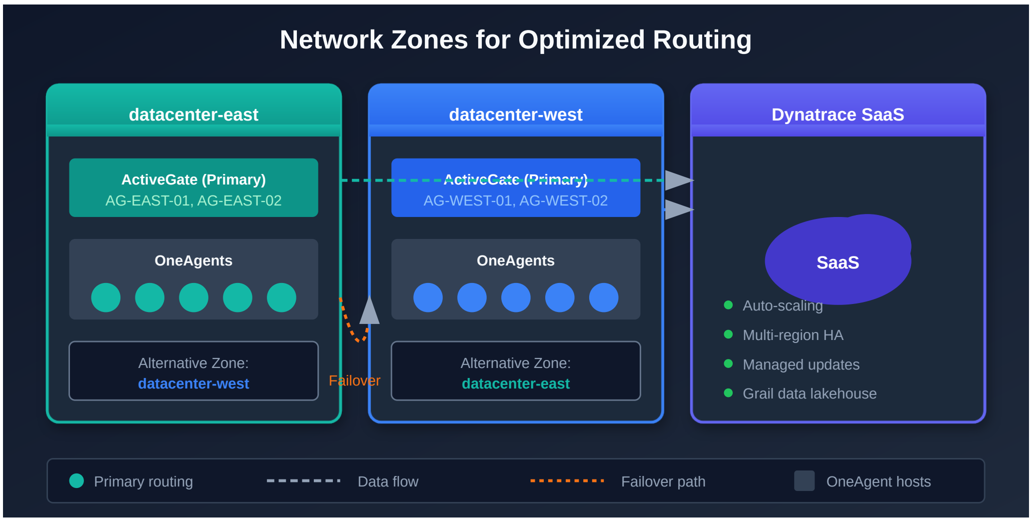
> **Tip:** Run these tests from multiple network segments to identify connectivity gaps before migration begins.

1.6 Proxy Configuration

If a proxy is required for outbound HTTPS:

```
# OneAgent proxy configuration
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
proxy=http://proxy.example.com:8080

# ActiveGate proxy configuration
# Edit /var/lib/dynatrace/gateway/config/custom.properties
# Add: proxy = http://proxy.example.com:8080
```

Proxy Consideration	Detail
SSL inspection	Must not break TLS to Dynatrace endpoints
Authentication	Supported (basic auth in proxy URL)
Allowlist	`*.dynatrace.com` on port 443



Zone Fallback Chain

Network zones follow a defined fallback hierarchy. Design your zones with this chain in mind:

Primary → Alternative → Default → Cross-zone fallback

Level	Behavior
Primary	Agent connects to AGs in its assigned zone
Alternative	If primary zone AGs are unavailable, falls back to the alternative zone
Default	If no alternative defined, falls back to the default zone
Cross-zone	Last resort — connects to any available AG across zones

Metadata-Driven Zone Assignment

Use host properties and metadata tagging to automate zone assignment:

Metadata Tag	Purpose	Example
`HOST_PROVIDER`	Cloud provider identification	`aws` , `azure` , `gcp`
`HOST_REGION`	Geographic region	`us-east-1` , `westeurope`
`HOST_ENVIRONMENT`	Environment tier	`production` , `staging`

> **Best Practice:** Leverage host properties to replace manual tags. This ensures consistent, automated zone assignment as new hosts are deployed.

2. Network Zones

Network Zones optimize routing of telemetry data from OneAgents to ActiveGates to SaaS. They are especially important in multi-datacenter or hybrid-cloud environments.

2.1 Why Network Zones Matter

Benefit	Description
Traffic optimization	Keep traffic local within network segments
Failover routing	Automatic failover to alternative zones
Bandwidth reduction	Minimize cross-datacenter WAN traffic
Logical grouping	Group agents and AGs by physical topology

2.2 Network Zone Planning

Create zones that match your physical network topology:

Zone Type	Naming Example	Description
Datacenter-based	`datacenter-east`	Primary datacenter in East region
Cloud region	`aws-us-east-1`	AWS region-specific zone
Environment-based	`production`	Production network segment
Hybrid	`dc-east-prod`	Datacenter + environment combined

2.3 Failover Configuration

Define alternative zones for each primary zone:

Primary Zone	Alternative Zone(s)	Failover Behavior
`datacenter-east`	`datacenter-west`	AGs in west serve east agents
`aws-us-east-1`	`aws-us-west-2`	Cross-region failover
`production`	`dr-production`	DR site serves production

2.4 Creating Network Zones

> **Sprint 1.339 deprecation (May 2026):** The dedicated `/api/v2/networkZones` Configuration API endpoint is deprecated. New automation should target the Settings 2.0 schema `builtin:networkzones.zones` via `POST /api/v2/settings/objects` — the same pattern used elsewhere in the AUTOM series. The legacy endpoint below still functions during the deprecation window so existing scripts keep working; treat the Settings 2.0 form as canonical for new work.

Settings 2.0 (recommended for new automation):

```
curl -X POST "https://{tenant}.live.dynatrace.com/api/v2/settings/objects" \
  -H "Authorization: Api-Token {token}" \
  -H "Content-Type: application/json" \
  -d '[{
    "schemaId": "builtin:networkzones.zones",
    "scope": "environment",
    "value": {
      "name": "datacenter-east",
      "description": "Primary datacenter in East region",
      "alternativeZones": ["datacenter-west"]
    }
  }]'
```

Legacy Configuration API (deprecated, still works during the deprecation window):

```
curl -X POST "https://{tenant}.live.dynatrace.com/api/v2/networkZones" \
  -H "Authorization: Api-Token {token}" \
  -H "Content-Type: application/json" \
  -d '{
    "id": "datacenter-east",
    "description": "Primary datacenter in East region",
    "alternativeZones": ["datacenter-west"]
  }'
```

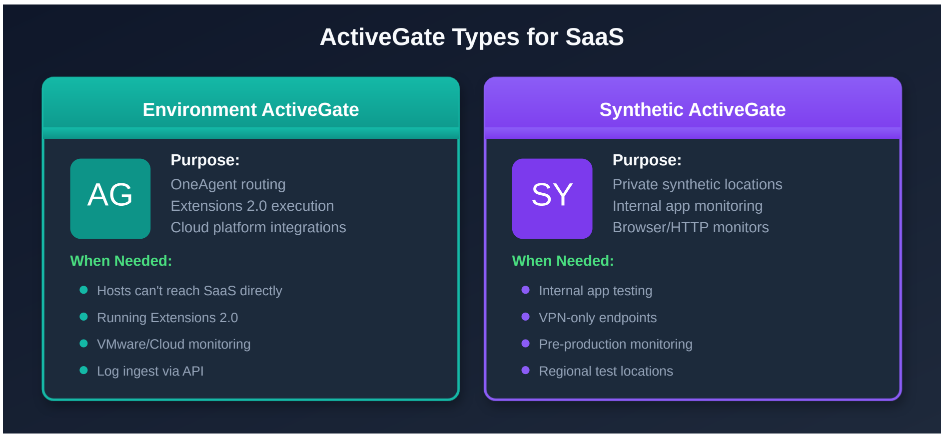
Assigning ActiveGates:

```
# During installation
sudo /bin/sh Dynatrace-ActiveGate-Linux.sh --set-network-zone=datacenter-east

# After installation
# Edit /var/lib/dynatrace/gateway/config/custom.properties
# Add: networkzone = datacenter-east
```

Assigning OneAgents:

```
# During installation
sudo /bin/sh Dynatrace-OneAgent-Linux.sh --set-network-zone=datacenter-east

# After installation
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-network-
zone=datacenter-east
```

> **Important:** Network Zones do NOT transfer from Managed—they must be recreated in your SaaS tenant before migrating OneAgents. Plan and deploy zones as part of Step 4 (Prepare).



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3. ActiveGate Design

3.1 Do You Need ActiveGates?

Scenario	ActiveGate Required?
Hosts can reach SaaS directly	No (optional for extensions)
Hosts behind firewall, no direct internet	**Yes**, for routing
Running Extensions 2.0	**Yes** (host-based AG required, not K8s-based)
Private synthetic monitoring	**Yes**
VMware monitoring	**Yes**

3.2 ActiveGate Sizing

Host Count (Routed)	CPU Cores	RAM	Disk
Up to 500	2	4 GB	20 GB
500–1,500	4	8 GB	40 GB
1,500–5,000	8	16 GB	80 GB
>5,000	Scale horizontally (multiple AGs)		

3.3 Placement Best Practices

Practice	Rationale
Close to monitored hosts	Minimize latency between OneAgents and AGs
Minimum 2 per network zone	High availability within each zone
Separate from monitored workloads	Avoid resource contention
In DMZ for cloud integrations	If needed for external access

3.4 ActiveGate Groups

Use AG groups to separate responsibilities:

Group Name	Purpose	Members
`production-routing`	Production OneAgent routing	AG-PROD-01, AG-PROD-02
`nonprod-routing`	Non-production routing	AG-NONPROD-01, AG-NONPROD-02
`extensions`	Extensions 2.0 execution	AG-EXT-01, AG-EXT-02
`synthetic`	Private synthetic locations	AG-SYNTH-01, AG-SYNTH-02

> **Note:** Extensions 2.0 require **host-based** ActiveGates. Kubernetes-based ActiveGates do not support Extensions 2.0.

3.5 Parallel Deployment Strategy

Install new SaaS-connected ActiveGates in parallel with existing Managed ActiveGates:

1. **Deploy new AGs** connected to the SaaS tenant
2. **Validate connectivity** from AGs to SaaS endpoints
3. **Redirect OneAgents** to new AGs during migration
4. **Decommission Managed AGs** after all agents are migrated

This approach maintains monitoring continuity and provides easy rollback if issues occur.

3.6 ActiveGate Footprint for Cloud-Native and Serverless Estates

The sizing table above is keyed on **routed host count** — the right model for a VM-heavy estate. A Cloud Run / GKE / OTLP-heavy estate routes very little through ActiveGates, so its AG footprint is minimal and driven by specific needs rather than by total workload count:

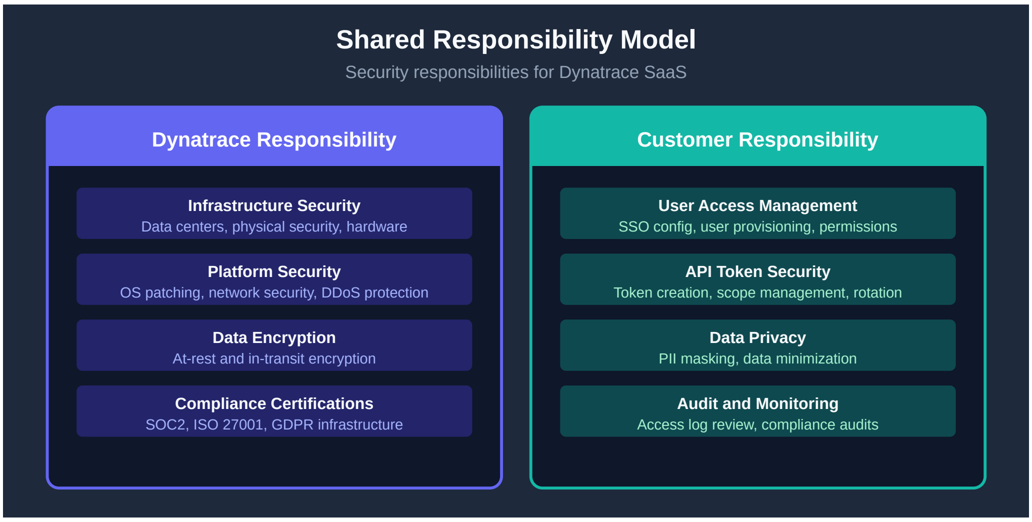
Cloud-native need	ActiveGate involvement
Cloud Run / OTLP apps → SaaS	Typically none — telemetry goes to the SaaS ingest/OTLP endpoints directly (or via an existing routing AG only if the network requires it)

Cloud-native need	ActiveGate involvement
GKE OneAgent (DynaKube) → SaaS	Direct, or via an in-cluster ActiveGate the Dynatrace Operator can deploy — not sized from the host table
GCP log forwarding	Runs as a container workload (Pub/Sub + forwarder), **not** an ActiveGate — see **CLOUD-06: GCP Integration** for forwarder sizing
Private synthetic monitoring	Dedicated synthetic ActiveGate(s) — sized by synthetic load, not routed hosts
Residual VMs / VMware / Extensions 2.0	Host-based ActiveGate(s) sized normally for just those remaining hosts

> **Rule of thumb:** For a predominantly cloud-native estate, size ActiveGates for the *residual* host/VM count plus any private-synthetic and extension needs — not for the total workload count. Most Cloud Run and OTLP traffic never touches an ActiveGate.

```
// Inventory current ActiveGates and their versions
smartscapeNodes "ACTIVEGATE"
| fields name, version = dt.active_gate.version, zone = dt.network_zone.id
| sort name asc

// Correction (verified 07/2026): this cell previously ran `fetch
dt.entity.active_gate` and
// carried a note claiming Smartscape had no ActiveGate node. Both were wrong.
There is NO classic
// ActiveGate entity type in any spelling (active_gate,
environment_active_gate,
// environment_activegate), so the classic query returned zero rows in every
tenant —
// indistinguishable from "no ActiveGates deployed". `smartscapeNodes
"ACTIVEGATE"` (no
// underscore) is the working path, and it works on tenants today.
// Field maps: entity.name → name, softwareVersion → dt.active_gate.version,
// networkZone → dt.network_zone.id.
```



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4. Security Architecture

4.1 SAML SSO Configuration

SaaS does not support LDAP—if your Managed environment uses LDAP authentication, you must migrate to SAML 2.0.

Authentication	Managed	SaaS
Local users	Cluster Management Console	Dynatrace Account Management
SAML/SSO	IdP → Managed cluster	IdP → Dynatrace Account
LDAP	Direct LDAP integration	**Not supported** (use SAML 2.0)

> **Critical:** Your Identity Provider (IdP) must sign the **entire SAML message**, not just the assertion. Failure to configure this correctly causes authentication errors. Azure Entra ID meets this requirement by default.

4.2 Azure Entra ID Considerations

Consideration	Requirement
SAML message signing	Full message (not just assertion)
Group claim limit	150 groups maximum per SAML token
Group filtering	**Filter claims to Dynatrace-related groups only**
Attribute mapping	Map UPN, email, first name, last name

> **Warning:** If a user belongs to more than 150 Azure Entra groups, the SAML token will contain a group overage claim instead of individual groups. Filter your SAML group claims to include only Dynatrace-relevant groups.

4.3 IAM Role Mapping

Map your existing Managed roles to SaaS IAM policies:

Managed Role	SaaS IAM Equivalent	Notes
Cluster administrator	Account administrator	Full account management
Environment administrator	Environment admin policy	Per-environment scope
Monitor user (read-only)	Viewer policy	Read-only access
Custom roles	Custom IAM policies	Recreate as policy statements

4.4 TLS and Encryption

Control	Detail
TLS version	TLS 1.2+ enforced for all connections
Data at rest	AES-256 encryption
Certificate pinning	OneAgent validates SaaS certificates
Per-tenant keys	Encryption keys isolated per tenant

4.5 Data Residency

Decision	Impact
Region selection	Choose at provisioning time (US, EU, APAC)
Cannot change later	Region is permanent after provisioning
Compliance alignment	Select region matching your regulatory requirements

> **Important:** Data residency region is selected during SaaS provisioning and **cannot be changed** after the fact. Confirm your region choice with compliance and legal teams before provisioning.

4.6 Firewall and Egress Controls

Control	Implementation
Egress filtering	Restrict outbound to `*.dynatrace.com` on port 443
IP allowlisting	SaaS endpoint IPs published by Dynatrace (subject to change)
DNS-based filtering	Preferred over IP-based (IPs may rotate)

4.7 API Token Strategy

Create purpose-specific tokens with minimal scopes:

Token Purpose	Required Scopes
OneAgent installation	`InstallerDownload`
ActiveGate installation	`InstallerDownload`
Configuration migration	`settings.read` , `settings.write`
Entity queries	`entities.read`
Metrics and logs	`metrics.read` , `logs.read`
AutomationEngine (Workflows)	OAuth client scopes `automation:workflows:read` , `automation:workflows:write` (not a classic API-token scope)

Bandwidth and Connectivity Planning

Consideration	Detail
Increased outbound bandwidth	Relocating the Dynatrace platform to SaaS increases outbound traffic from your network — plan for this with network team
SSL certificates	SSL certificates are possible for ActiveGates but NOT for the Dynatrace SaaS UI
Service requests	Plan for change management procedures, change windows, and deployment freezes
OneAgent re-homing strategy	Choose one: (1) Uninstall agents and install new, or (2) Update agents in place via `oneagentctl` to keep IDs (host, process group, services) and metadata

5. High Availability Design

5.1 SaaS-Side HA (Provided by Dynatrace)

Dynatrace SaaS includes built-in high availability:

Capability	Detail
Multi-region deployment	Automatic geographic redundancy
Automatic failover	No customer action required
Data replication	Cross-region data copies
SLA	99.5%+ availability (varies by contract)

5.2 Customer-Side HA

Your responsibility is ensuring no single points of failure in the agent-to-SaaS path:

Component	HA Strategy
ActiveGates	Minimum 2 per network zone

Component	HA Strategy
Network zones	Define alternative zones for failover
Synthetic AGs	Deploy pairs at each private location
Load balancing	OneAgent auto-discovers available AGs (no LB needed)

5.3 Failover Behavior

Scenario	OneAgent Behavior
Primary AG unavailable	Automatically fails over to secondary AG in same zone
All AGs in zone unavailable	Fails over to AGs in alternative network zone
All AGs unavailable	Buffers data locally (up to 2 hours)
AG + SaaS restored	Resumes normal transmission, flushes buffer

5.4 Design for Zero Single Points of Failure

Layer	Requirement
Network zones	At least one alternative zone per primary
ActiveGates per zone	Minimum 2 per zone
Network paths	Redundant paths from agents to AGs
DNS	Multiple DNS resolvers for SaaS domains

```
// Verify ActiveGate distribution across network zones
smartscapeNodes "ACTIVEGATE"
| summarize agCount = count(), by:{dt.network_zone.id}
| sort agCount desc

// Correction (verified 07/2026): this cell previously ran `fetch
dt.entity.active_gate` and
// carried a note claiming Smartscape had no ActiveGate node. Both were wrong.
There is NO classic
// ActiveGate entity type in any spelling (active_gate,
environment_active_gate,
// environment_activegate), so the classic query returned zero rows in every
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// indistinguishable from "no ActiveGates deployed". `smartscapeNodes
"ACTIVEGATE"` (no
// underscore) is the working path, and it works on tenants today.
// Field maps: entity.name → name, softwareVersion → dt.active_gate.version,
// networkZone → dt.network_zone.id.
```

6. Architecture Design Checklist

Complete this checklist before proceeding to Step 4 (Prepare).

Network

Checkpoint	Status
Network firewall rules documented (outbound 443 to SaaS)	[ ]
DNS resolution verified for `{tenant-id}.live.dynatrace.com`	[ ]
DNS resolution verified for `{tenant-id}.apps.dynatrace.com`	[ ]
Connectivity tested from representative hosts	[ ]
Proxy configuration documented (if required)	[ ]

Network Zones

Checkpoint	Status
Network zone topology designed (matching physical topology)	[ ]

Checkpoint	Status
Alternative zones defined for failover	[ ]
Zone naming convention established	[ ]

ActiveGates

Checkpoint	Status
ActiveGate sizing calculated based on host counts	[ ]
Placement planned (close to monitored hosts)	[ ]
Minimum 2 AGs per network zone for HA	[ ]
AG groups defined (routing, extensions, synthetic)	[ ]
Host-based AGs planned for Extensions 2.0 (if needed)	[ ]

Security

Checkpoint	Status
SSO/SAML configuration designed	[ ]
IdP configured to sign full SAML message (not just assertion)	[ ]
Azure Entra group claims filtered to Dynatrace groups	[ ]
IAM policy mapping complete (Managed roles → SaaS policies)	[ ]
API token strategy defined with minimal scopes	[ ]
Data residency region confirmed with compliance team	[ ]
TLS 1.2+ confirmed for all connections	[ ]

High Availability

Checkpoint	Status
No single points of failure in AG topology	[ ]
Network zone failover paths verified	[ ]

Checkpoint	Status
SaaS SLA reviewed and documented	[ ]

Next Step

> → **M2S-04: Step 4 — Prepare** — Get your SaaS environment ready for migration execution. Deploy ActiveGates, configure network zones, create API tokens, and install the SaaS Upgrade Assistant.

Continue the Series

Next Notebook	Focus
M2S-04: Step 4 — Prepare	Environment setup and migration preparation

Architecture Resources

- ActiveGate Documentation
- Network Zones
- OneAgent Network Requirements
- IAM Documentation
- Dynatrace Trust Center
- SaaS Upgrade Assistant

Summary

In this notebook, you designed:

- **Network architecture** — Firewall rules, DNS, proxy configuration, and connectivity testing

- **Network zone topology** — Zones matching physical topology with failover alternatives
- **ActiveGate deployment** — Sizing, placement, groups, and parallel deployment strategy
- **Security architecture** — SAML SSO, IAM role mapping, TLS, data residency, and token strategy
- **High availability** — Redundant AGs, zone failover, and zero single points of failure

> **Key Takeaway:** The primary architectural change is network connectivity. Design your network zones and ActiveGate topology before migration begins—these decisions are difficult to change after OneAgents are redirected. Confirm data residency and SSO configuration with your compliance and security teams early.

Continue to M2S-04: Step 4 — Prepare to set up your SaaS environment for migration execution.
